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PoolParty: streamlined design of DNA sequence libraries in Python

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Abstract

Background Computationally designed DNA sequence libraries are essential components of massively parallel reporter assays (MPRAs), deep mutational scanning (DMS) experiments, and other multiplex assays of variant effect (MAVEs). They are also increasingly used in silico to analyze genomic AI models. Designing these libraries, however, remains tedious and error-prone due to the scarcity of purpose-built software.

Results Here we describe PoolParty, a Python package that streamlines the design of complex oligo pools using a simple but flexible API. In PoolParty, each library is represented by a computational graph that can be specified in just a few lines of code. Over 50 built-in operations cover nucleotide- and codon-level mutagenesis, motif insertion, barcode generation, and more. PoolParty automatically generates informative names for each sequence and provides "design cards" detailing how each sequence was generated. Visualization methods let users quickly audit library content and inspect the underlying graph. PoolParty thus transforms oligo pool design from a tedious task requiring custom functions and scripts into a structured, transparent, and reproducible process.

Conclusions PoolParty streamlines the design of DMS, MPRA, and other multiplex assay libraries, and the design cards it provides can help researchers systematically probe and interpret genomic AI models. PoolParty can also be extended to support new assays and analysis strategies as they emerge.

Keywords Sequence design, Library design, Massively parallel reporter assay, Deep mutational scanning, Multiplex assay of variant effect, Genomic AI, In silico experiments, Python software, Directed acyclic graph

Background

Designed libraries of DNA sequences (oligonucleotide pools) have become essential tools in functional genomics. In massively parallel reporter assays (MPRAs), complex libraries of DNA sequences are routinely used to measure the regulatory activities of tens of thousands of candidate regulatory elements or their variants in a single experiment [1–13]. In deep mutational scanning (DMS) experiments, coding sequences that have been either systematically or randomly mutagenized are commonly used to map

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protein fitness landscapes [14–20]. Oligo pools can also be used in high-throughput biochemical assays, such as measurements of transcription factor specificity [21–27]. These experiments are all examples of multiplex assays of variant effect (MAVEs), a large and expanding family of high-throughput methods that quantitatively link sequence to function at scale [28–30]. Beyond the wet lab, designed sequence libraries are increasingly used in silico to probe the behavior of genomic AI models [31–35] and to augment genomic AI training data [36].

Computationally designing these libraries is straightforward in principle, but in practice it can be tedious and error-prone. Typical applications require oligo pools comprising multiple types of variant sequences. A DMS library, for example, might combine exhaustive single-amino-acid substitutions, sampled higher-order mutants, multiple copies of wild-type and control sequences, and barcodes—each generated by a different procedure and subject to different coverage requirements. An MPRA library might tile multiple transcription factor binding sites across a regulatory element in all possible permutations and orientations, then assign each variant multiple barcodes. Coordinating this kind of combinatorial logic in a one-off script quickly becomes unwieldy, and often produces errors that are hard to catch.

Existing tools can help with parts of this process. General-purpose sequence toolkits [37–39] can manipulate sequences but have no notion of a variant library as a structured object. Consequently, users must write custom scripts to handle the combinatorial logic themselves. Several other tools automate library design or sequence perturbation for particular applications. For example, VaLiAnT [40] generates mutational scanning libraries of both coding and noncoding regions; supported mutation types include single-nucleotide substitutions, amino acid substitutions, and in-frame deletions. For MPRAs, MPRAnator [41] provides modules for motif placement, SNP design, and negative control generation (see also [42]). For in silico experiments, tangermeme [43] supports sequence perturbations and the analysis of genomic deep learning models. However, each of these tools is targeted to specific applications and, as a result, no individual tool can generate the full range of library designs in common use. For example, VaLiAnT does not support regulatory sequence libraries containing random combinations of transcription factor binding sites, while MPRAnator and tangermeme do not support codon-aware mutagenesis of protein-coding sequences.

Here we present PoolParty, a Python package that enables the streamlined design of complex DNA sequence libraries. In PoolParty, libraries are represented as objects called *Pools*, while individual steps in the sequence generation process are called *Operations*. Using a concise and transparent syntax, users chain together Operations into a directed acyclic graph (DAG)—an approach inspired by deep learning frameworks—that yields the desired Pool. The DAG describes the high-level logic used to generate sequences, while the procedural details and bookkeeping are handled internally. Importantly, no sequences are generated until the user requests them, and it is simple to generate small test sets of sequences. This allows users to explore multiple design options without having to generate complete libraries at each iteration. When sequences are generated, each is automatically assigned an informative name and paired with a *design card* that records the specific procedures used to construct it. We demonstrate PoolParty through three example applications: a DMS library for protein GB1, an MPRA library probing transcriptional regulatory grammar, and an in silico experiment characterizing the genomic

deep learning model SpliceAI [44], in which design cards provide the covariates for surrogate modeling.

Implementation

Pools and Operations

PoolParty is built around two core abstractions: Pools and Operations. Pools represent collections of sequences—either the final library that the user wishes to generate, or intermediate sets of sequences out of which the final library is constructed. Operations represent the steps used to create Pools. When designing a library, users chain together multiple Operations into a DAG that yields the desired Pool, called the *root Pool*. Sequences are then generated on demand from the root Pool. By delaying sequence generation until it is explicitly requested, users can test many different library designs before committing to generating the full library. For example, users can inspect a few sampled sequences from each one of multiple proposed designs.

PoolParty provides over 50 built-in Operations that users can choose from when constructing DAGs. Each Operation takes zero or more Pools as input and returns exactly one Pool as output. It is useful to think of Operations as falling into four functional categories (Fig. 1A): source, transformation, composition, and state Operations. *Source Operations* create the initial Pools from which all downstream Pools are constructed. The Pools they create may represent user-specified sequences loaded directly from

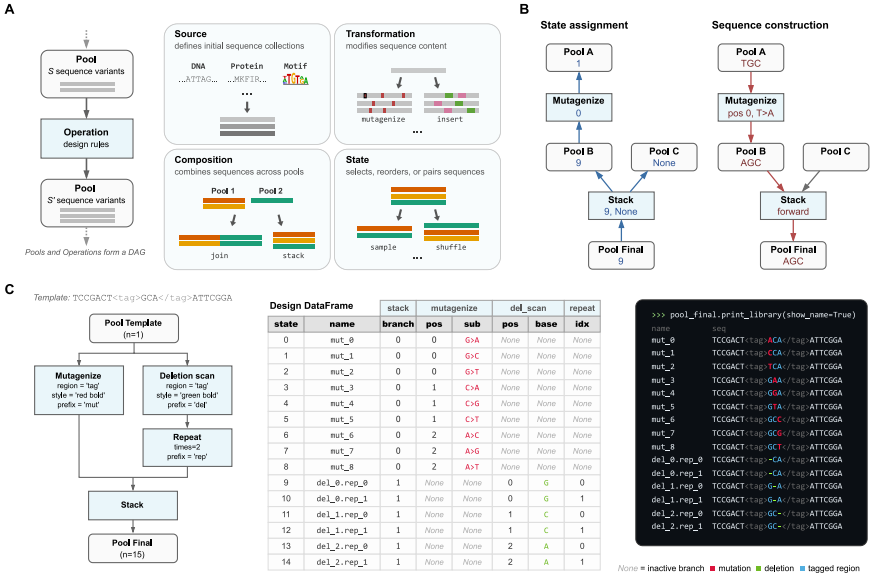


Fig. 1 PoolParty builds libraries from Pools and Operations. **A** Pools represent collections of sequences, and Operations generate new Pools based on their inputs (left). Users chain multiple Operations into a directed acyclic graph (DAG) that specifies the desired library. Operations roughly fall into four functional categories: source, transformation, composition, and state (right). **B** Sequence generation proceeds in two steps: *state assignment* and *sequence construction*. For each desired sequence, PoolParty passes to the root Pool a number (called the *state*) that specifies which sequence to generate. During state assignment (left), this state is propagated backward through the DAG, resolving the internal state of every Operation along the way. During sequence construction (right), each Operation generates and combines sequence fragments according to its assigned state, producing the final sequence at the root Pool. When multiple Pools are combined by `stack`, only one parent is active for a given state; inactive branches receive no state (`None`) and are shown as gray arrows. This example is expanded on in Supplemental Information, Fig. S1. **C** Example workflow. The template sequence contains a user-specified region (called "tag") that is targeted for both mutagenesis and deletion scanning. Left: the DAG specifying the sequence library. Center: the design card DataFrame returned by the DAG. Right: styled sequence output, with colors indicating mutations (red), deletions (green), and the tag region (blue)

FASTA or VCF files, DNA sequences generated using position weight matrices (PWMs) or IUPAC motifs, random k-mers, barcodes, and so on. *Transformation Operations* modify sequence content, e.g., generating sequence variants through point mutations, insertions, deletions, recombination, or shuffling. ORF-aware Operations are provided for transformations that operate on open reading frames. *Composition Operations* combine sequences from multiple parent Pools, either by concatenating parent sequences end-to-end (the `join` Operation) or by merging multiple Pools into one Pool (the `stack` Operation). *State Operations* affect which sequences are kept in a Pool and how they are ordered, and include methods to select, reorder, filter, and replicate sequences. Sequences can be filtered based on sequence length, GC content, homopolymer length, sequence complexity, restriction enzyme recognition sites, or other user-defined criteria.

Sequence generation

To generate a sequence library, PoolParty iterates through a series of *states*. Each state is an integer that, together with a random seed, uniquely determines a sequence. The generation of each individual sequence proceeds in two steps: *state assignment* and *sequence construction* (Fig. 1B; Supplemental Information, Fig. S1). This process begins when the corresponding state is passed to the root Pool. State assignment involves a *backward pass* through the DAG, from the root Pool toward the source Pools. Each Pool passes the state it receives to the Operation that created it. Each Operation then decomposes this state into an *internal state* (which it keeps) and zero or more *component states*, each of which is passed to a distinct input Pool of that Operation. Sequence construction then occurs during a *forward pass* through the DAG, from the initial Pools toward the root Pool: each Operation constructs a sequence based on its internal state and the sequences returned by its input Pools, then passes this sequence to the Pool it creates. The final constructed sequence is then received from the root Pool and recorded.

Operation modes

The specific way that each Operation interprets its internal state is determined by its *mode*.

- In *sequential mode*, the Operation exhaustively enumerates a finite set of sequence designs. For example, when `mutagenize` is instructed to generate single-nucleotide mutations in sequential mode, it yields one variant for each possible character change at each target position. Its internal state specifies which mutation to apply at which position.
- In *random mode*, the Operation samples sequence designs stochastically. For example, when `mutagenize` operates in random mode, it randomly selects which positions to mutate and which mutations to introduce. Its internal state, combined with a master seed that determines the specific random draws, ensures reproducibility.
- In *fixed mode*, there is no internal state. Rather, output sequences are uniquely determined by the input sequences. One example of a fixed-mode operation is `join`, which concatenates its input sequences without modification.

StateTracker

The way each Operation decomposes the state it receives during the backward pass is designed to reverse the way it constructs sequences during the forward pass. For most Operations, the set of possible output states is the Cartesian product of the possible internal states of the Operation together with the possible states of each input Pool. In such cases, the Operation uses mixed-radix decomposition to split the received state into its component states. The `stack` Operation is different: its output states are the disjoint union of its input Pool states. The `stack` Operation therefore uses the state it receives to select which input Pool to use, and to instruct that Pool (but not the other Pools) which sequence to return. Keeping track of these rules across a complex DAG requires specialized bookkeeping. We therefore developed a back-end package called StateTracker that automates state composition and decomposition during forward and backward passes through the DAG.

Sequence regions

One may want to perform different Operations on different parts of a sequence. To facilitate this, PoolParty lets users mark regions of interest within sequences using XML-style tags. PoolParty supports both opening/closing tag pairs (e.g., `AAA<cre>CCCGGG</cre>TTT`) and self-closing tags (e.g., `ACGT<ins/>ACGT`). Operations can then be instructed to act only on a specified region, as illustrated in Fig. 1C. Tags persist through the DAG and remain valid even when upstream Operations change the content of a region. Multiple Operations can thus target modifications to the same region in series.

Sequence metadata

It is often useful to record how each sequence within a library was constructed. PoolParty provides three complementary ways to do this (Fig. 1C).

- *Design cards* record the specific changes applied to each sequence by each Operation. Users specify which Operations and Operation-specific variables to track, including user-defined sequence scores computed with the `score` Operation. As each sequence is generated, the root Pool returns it together with a design card that records the tracked variable values for that sequence. PoolParty compiles these into a DataFrame in which each row contains a generated sequence and its design card.
- *Sequence names* summarize at a glance how each sequence was constructed. Users assign a prefix to each Operation they wish to track. PoolParty then pairs each prefix with the corresponding Operation's internal state to form a token (e.g., `mut_03`), and concatenates these tokens to form the sequence's name. For example, `wt.mut_03.bc_01` might denote a wild-type sequence with mutation variant number 3 and barcode number 1.
- *Sequence styling* allows sequences to be annotated with colors and formatted text that reflect each one's construction history. For example, `mutagenize` can be instructed to mark the positions it mutates in bold yellow text, making these positions stand out from the rest of the sequence. This formatting is accomplished using ANSI escape codes, so that output renders correctly in command-line terminals, Jupyter notebooks, and other environments that support basic text formatting. These styles combine in a natural manner as sequences propagate through the DAG. For example,

a mutated position can be underlined even if it occurs within a sequence region that is already colored blue.

PoolParty also provides a `stats` method for summarizing a DNA Pool, including its unique and duplicate sequence counts, pairwise Hamming distances, and homopolymer frequencies.

Implementation and extensibility

PoolParty requires Python 3.10 or later, is designed to have minimal dependencies, and is distributed via PyPI. Documentation, tutorials, and example notebooks are available on the ReadTheDocs and GitHub project websites. PoolParty is also extensible: users can create new Operation types by subclassing the Operation base class and implementing a few required methods. Custom Operations automatically inherit support for state tracking, design card generation, and visualization, allowing researchers to add domain-specific functionality without reimplementing core infrastructure.

SpliceAI surrogate modeling analysis

This section provides technical details for the surrogate modeling analysis presented in Results. Using the human 5' splice site (5'ss) position weight matrix [45], we sampled 9-mers representing cryptic 5'ss sequences and scored them with MaxEntScan [46]. Sequences were binned into 100 quantiles of MaxEntScan score (hereafter, strength), and 20 sequences were drawn from each bin (2,000 total). Each 9-mer was inserted at 100 positions flanking the canonical 5'ss of SMN2 exon 7 and paired with a matched control in which the GT dinucleotide was disrupted (T > A at + 2).

SpliceAI [44] is a genomic deep learning model for splice-site prediction in humans. We used the published five-model ensemble to predict the canonical 5'ss probability using ± 5 kb of genomic context (GRCh38). For each library-control pair, we computed the difference in logit-transformed predictions at the canonical 5'ss. These per-sequence values were averaged within each cell of the resulting 100 × 100 position-by-strength grid.

We fit generalized additive models (GAMs) to the cell means using pyGAM [47], with smooth terms for position, strength, and their interaction. Models were weighted by the inverse variance of each cell mean. Exonic and intronic regions were modeled separately. Model performance was evaluated by checkerboard cross-validation of the grid.

Use of large language models

The authors used large language models (primarily Claude Opus 4.5 and 4.6, Anthropic) to assist with code development, debugging, and documentation writing. The manuscript was written entirely by the authors, with LLM assistance only at the copy-editing stage. The figures were also prepared by the authors; LLMs were used only to help write the scripts that generate the plots. The authors affirm that they are fully responsible for the content of the manuscript, the codebase, and the documentation.

Results

Example 1: DMS library for protein GB1

As a first example, we use PoolParty to reconstruct and extend the library used in a DMS experiment on protein GB1 by Olson et al. [17]. Their library contained the wild-type

GB1 coding sequence, all single-amino-acid substitutions (excluding the start codon), and all possible pairs of amino acid substitutions. This library is substantially larger than most DMS libraries, containing over half a million variants, and has become a benchmark dataset in the quantitative modeling of protein sequence-function relationships [48–51]. To the Olson et al. library we added random higher-order amino acid substitutions as well as 100 replicates of the wild-type sequence.

Fig. 2 A illustrates the DAG used to construct the resulting library; the corresponding Python code is shown in Fig. 2B. First, we use the `from_seq` Operation to define a Pool named `orf_pool` that represents the wild-type GB1 coding sequence. This Pool is then passed to four separate Operations that define the four library components. The first is a `repeat` Operation that produces 100 copies of the wild-type sequence. The second and third are `mutagenize_orf` Operations running in sequential mode: one generates all 1,045 single-amino-acid substitutions, and the other generates all 536,085 pairwise amino acid substitutions. The fourth is a `mutagenize_orf` Operation running in random mode, which mutates each amino acid position with 10% probability to generate 10,000 higher-order variants. We note that `mutagenize_orf` supports multiple codon mutation types; this example uses `missense_only_first`, which selects the codon most frequently used in the human genome for each target amino acid. These four Pools are then merged by `stack` into the root Pool, yielding a library of 547,230 sequences in total. The design card for each variant (Fig. 2C) reports which component it belongs to, the specific mutations it contains, and the replicate index (for wild-type copies). Users



Fig. 2 A deep mutational scanning (DMS) library for protein GB1. **A** DAG and representative sequences for each of the four library components: replicates of the wild-type sequence, all single-amino-acid substitutions, all pairwise amino acid substitutions, and random higher-order mutants. Mutated codons are highlighted with amino acid translations shown beneath. **B** Python code implementing the DAG. **C** Design cards reporting the library component, mutation positions, and amino acid changes for representative variants. **D** DAG structure displayed by `print_dag()`

can also inspect the DAG structure by calling the `print_dag` method of the root Pool (Fig. 2D).

Example 2: MPRA library for regulatory grammar

MPRA libraries used to study regulatory grammar often vary the positions and orientations of transcription factor binding sites (TFBSs) [3, 5, 52–56]. For example, Georgakopoulos-Soares et al. [56] studied the impact of order and orientation on gene expression by testing over 200,000 sequences containing all pairwise and triplet arrangements of 18 TFBSs. Designing such libraries requires substantially more complex combinatorics than standard DMS libraries do. PoolParty handles this complexity without making additional demands on the user.

As a second example application of PoolParty, we designed an MPRA library representing various arrangements of binding sites for three liver-enriched transcription factors—HNF4A, PPARA, and XBP1 (Fig. 3A, B). Following the oligo layout of Melnikov et al. [3], we start by creating a Pool representing a template sequence that contains a putatively inert 100 bp cis-regulatory element (CRE) region (`<cre>`) and a barcode region (`<bc>`). Next, we create a Pool representing the binding site for HNF4A. This Pool is then passed to the `flip` Operation, which yields both orientations of the site (forward and reverse complement). Pools for PPARA and XBP1 are created similarly. The template Pool and three TFBS Pools are then passed to the `insertion_multiscan` Operation, which samples 1,000 random placements of the TFBSs within the `<cre>` region without overlap. Because `flip` runs in sequential mode, all $2^3 = 8$ orientation combinations are generated for each placement, yielding 8,000 unique CRE variants. Each sequence is

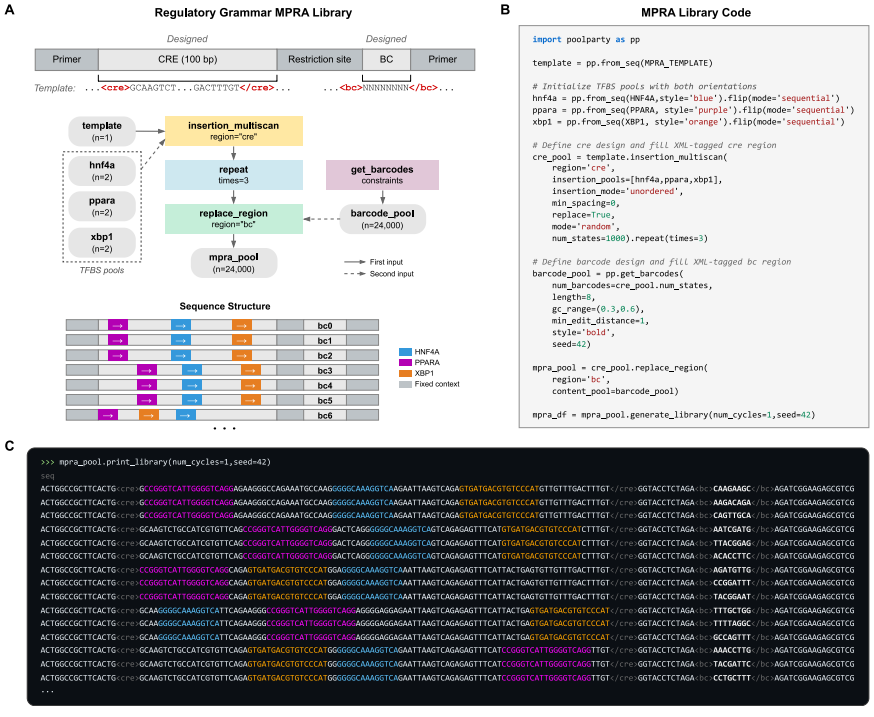


Fig. 3 A massively parallel reporter assay (MPRA) library for probing regulatory grammar. **A** Library schematic showing TFBS placements across a 100-bp CRE in representative variants, each linked to three different barcodes. **B** Python code for library construction. TFBS Pools are initialized with forward sequences and augmented with the `flip` Operation to include both orientations. **C** Output of `print_library`. TFBSs are colored by identity (HNF4A blue, PPARA purple, XBP1 orange); barcodes are shown in bold

then replicated three times using the `repeat` Operation. Finally, the `get_barcodes` Operation is used to generate unique barcode sequences that are inserted into the `<bc>` region of the template. The result is a library of 24,000 sequences (Fig. 3A).

The text styling capabilities of PoolParty help users verify the architecture of the resulting sequences. After the template Pool is created, the `stylize` Operation is used to shade the `<cre>` region gray. Each TFBS Pool is also assigned a distinct color when it is created (HNF4A blue, PPARA purple, XBP1 orange); this is specified by the `style` parameter passed to `from_seq`. Barcodes are similarly set to render in bold text (Fig. 3B). These styles compose as sequences pass through the DAG, so that when `print_library` is called (Fig. 3C), the position, order, and orientation of each binding site are immediately visible.

Example 3: surrogate modeling of SpliceAI predictions

Designed sequence libraries are also used to study genomic AI models [31–33, 35]. First, an in silico experiment is carried out in which the AI model scores all sequences in a library. The results are then analyzed to characterize model behavior. One useful way of analyzing these synthetic data is *surrogate modeling*, in which a mathematically simple model—one whose structure and parameters are directly interpretable—is fit to the predictions.

In addition to streamlining the design of sequence libraries, PoolParty facilitates in silico experiments on genomic AI models in two ways. First, because sequence design is separated from sequence generation, new sequences can be generated on demand as the experiment proceeds. Second, the design card that PoolParty returns with each sequence provides metadata that can be used directly as covariates in downstream analyses, including surrogate modeling.

As a third example application of PoolParty, we carried out a surrogate modeling study of SpliceAI, a deep learning model of mRNA splicing. We asked how the insertion of a cryptic 5'ss near an existing (canonical) 5'ss affects the model's prediction for canonical site usage. Using the 5'ss of SMN2 exon 7 as the canonical site, we used PoolParty to construct a library in which cryptic 5'ss 9-mers of varying strength (i.e., MaxEntScan score [46]) were inserted at varying positions on either side of the canonical 5'ss (see Implementation). The specific values for cryptic site strength and position were recorded in the design cards (Fig. 4A).

Each variant sequence was paired with a matched control in which the cryptic site was disrupted, and the effect of each insertion was quantified as the difference in logit-transformed SpliceAI-predicted probabilities at the canonical site (Δlogit). Generalized additive models (GAMs) were used to separately model the effects of cryptic site insertion at exonic and intronic positions. Each of these two models has three terms, corresponding to the effects of position alone, strength alone, and interactions between these covariates (Fig. 4A).

The GAMs closely reproduced the observed data (Fig. 4B, C). Cryptic sites on the exonic side suppressed canonical site usage more strongly than those on the intronic side (Fig. 4D), with the effect peaking approximately 20 bp upstream of the canonical 5'ss. This is consistent with the observation that cryptic 5'ss are preferentially depleted near annotated ones on the exonic side [57]. Stronger cryptic sites drove greater suppression in both models (Fig. 4E). The effects of position and strength exhibited substantial

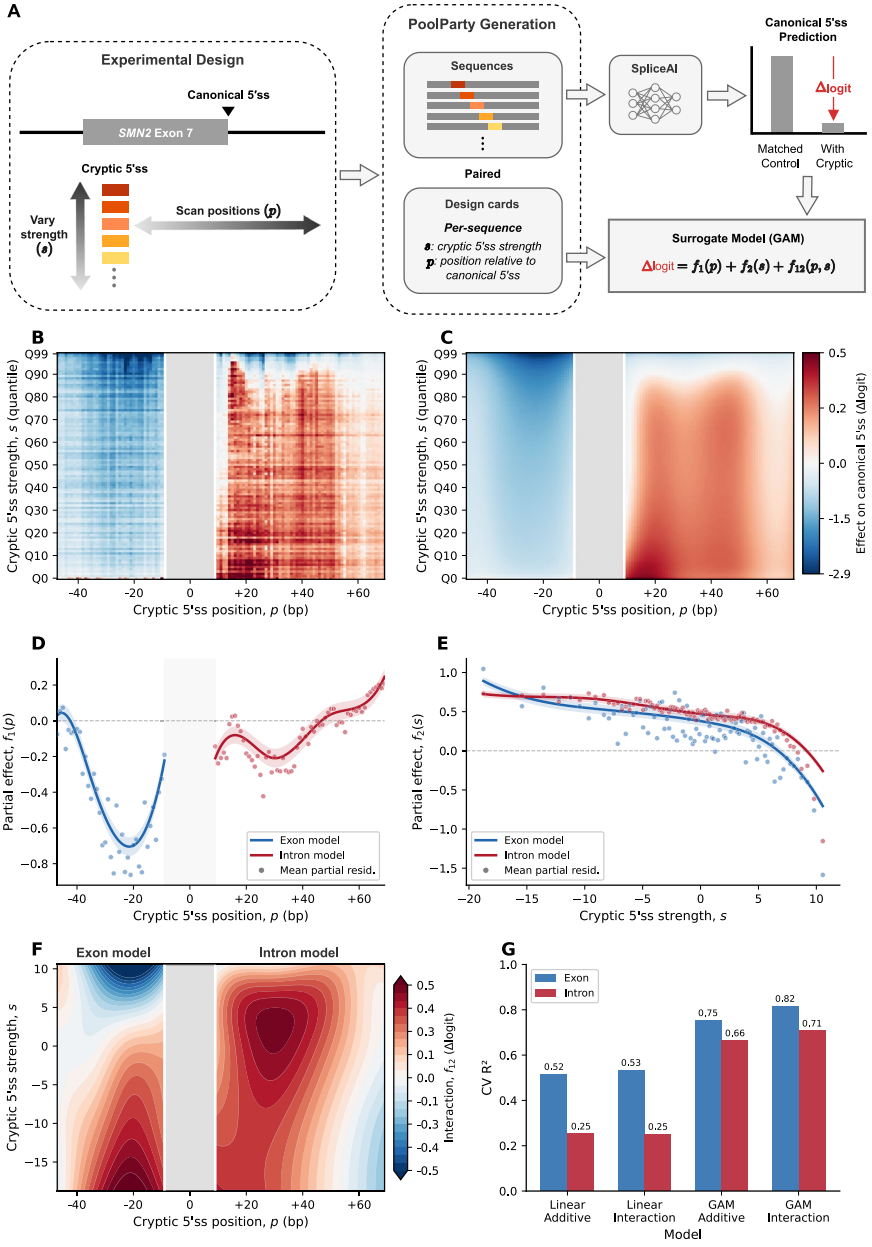


Fig. 4 Surrogate modeling of SpliceAI predictions. **A** Experimental design. Cryptic 5′ss 9-mers of varying MaxEntScan strength (*s*) were inserted at varying positions (*p*) flanking the canonical 5′ss of SMN2 exon 7. Each variant sequence was paired with a control sequence in which the splice-site GT dinucleotide is disrupted (GT → GA). PoolParty records the values of *s* and *p* in the design cards. The effect of each insertion on SpliceAI predictions at the canonical 5′ss is quantified as Δlogit and modeled as a function of *s* and *p* using a generalized additive model (GAM). **B** Mean Δlogit across the strength × position grid (100 × 100; each cell averages over 20 distinct 9-mers). The gray band marks the canonical 5′ss region. **C** GAM predictions (including the interaction term), using the same color scale as in B. **D** Partial dependence of the position term $f_1(p)$, shown separately for exon (blue) and intron (red) models. Shaded regions indicate 95% confidence intervals; points denote mean partial residuals. **E** Partial dependence of the strength term $f_2(s)$, with conventions as in D. **F** Interaction surface $f_{12}(p, s)$ for exon and intron models. **G** Checkerboard cross-validation R^2 for linear and GAM models, with and without interaction terms

interactions (Fig. 4F). Through a comparison to analogous linear models, we observed that the non-linear terms and interaction terms in the GAMs substantially improved predictive performance, with held-out R^2 values of 0.82 and 0.71 for exonic and intronic insertions, respectively (Fig. 4G).

Discussion

We have presented PoolParty, a Python package for designing oligonucleotide libraries. PoolParty facilitates this task by representing libraries as directed acyclic graphs (DAGs) that users can construct using a simple API. We demonstrated PoolParty through three use cases: a DMS library for protein GB1; an MPRA library probing transcriptional regulatory grammar; and a surrogate modeling analysis of the effects of cryptic 5'ss on SpliceAI predictions. These examples illustrate the flexibility of PoolParty and its applicability to both experimental and computational studies. Runtime and peak memory benchmarks for the three examples are provided in Supplemental Information (Fig. S2 and Table S1).

PoolParty was inspired by deep learning frameworks like PyTorch and TensorFlow, in which a high-level API is used to construct a DAG representing the desired neural network. The DAG then enables computation of model gradients via backpropagation. Similarly, in PoolParty, a backward pass through the DAG is used to decompose a state specific to each requested sequence into the specific Operations needed to construct that sequence during a subsequent forward pass through the DAG. This state-decomposition mechanism is a core innovation of PoolParty; in particular, it makes sequence generation random-access and allows users to test complex oligo pool designs without having to generate full sequence libraries.

We also highlighted the design card that PoolParty provides with each sequence it generates. Among other things, design cards facilitate using the parameters governing each sequence's construction as covariates in downstream analyses. Design cards are thus a bookkeeping device that allows users to avoid reverse-engineering sequence architecture from raw sequence when investigating how this architecture affects the output of a sequence-to-function model. For example, in the SpliceAI analysis of Fig. 4, the cryptic splice-site strength and position recorded in each design card provided the covariates used for surrogate modeling. We expect design cards to be increasingly useful as in silico experiments on genomic AI models become more common.

PoolParty complements existing library-design tools such as VaLiAnT [40], MPRAnator [41], and tangermeme [43] by providing a single framework for DMS libraries, MPRA libraries, and library designs that combine elements of both. Its DAG-based structuring of oligo pools lets users freely compose arbitrary sequence generation and transformation operations. Because the DAG encodes the full construction logic, structured design-card metadata are generated automatically as a byproduct. Additional comparisons between PoolParty and existing tools are provided in Supplemental Information, Figs. S3 and S4 and Tables S2 and S3.

PoolParty was designed to solve a specific computational problem: streamlining the design of DNA sequence libraries for a wide range of experimental applications and in silico studies. As such, it has several limitations. First, PoolParty operates on sequences, not genomic loci or transcript models. It can accept genomic coordinates as input to extract sequences from FASTA files, but does not interpret transcript annotations (e.g.,

in GFF/GTF files) or keep track of genomic coordinates through subsequent design steps. This may make PoolParty less helpful for certain genome-editing applications (e.g., as compared to VaLiAnT). PoolParty is also not a sequence optimization tool; it does not design sequences that have optimized codon usage [58], that are subject to complex synthesis constraints [59], or that have optimal activity as assessed by machine learning models [60]. PoolParty also does not address the problem of physically constructing DNA sequence libraries, e.g., by designing mutagenic primers [61]. We expect, however, that PoolParty can in many cases be productively paired with complementary tools that do address these challenges.

Conclusions

PoolParty streamlines the design of DMS experiments, MPRAs, and other multiplex assays of variant effect. As genomic AI models continue to advance, we expect that PoolParty will help researchers systematically probe and interpret the behavior of these models. Moreover, because PoolParty's composable design makes it straightforward to define new Operations, the framework can readily be extended to support novel assay types and analysis strategies as they emerge.

Abbreviations

5'ss	5' splice site
CRE	Cis-regulatory element
DAG	Directed acyclic graph
DMS	Deep mutational scanning
GAM	Generalized additive model
MAVE	Multiplex assay of variant effect
MPRA	Massively parallel reporter assay
PWM	Position weight matrix
TFBS	Transcription factor binding site

Supplementary information

The online version contains supplementary material available at https://doi.org/10.1186/s12859-026-06696-6.

Supplemental Information: Supplemental Methods, Figs. S1–S4, Tables S1–S3.

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Author contributions

Z.L. conceived the project. Z.L. and J.B.K. designed the project, wrote the software, performed the analyses, and wrote the manuscript. Z.L. and A.C. wrote the documentation, developed the test suite, and deployed the software. J.B.K. supervised the research. All authors read and approved the final manuscript.

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Data availability

No experimental data were generated or analyzed in this study. The sequence and software used in the paper were obtained from public sources referenced in the manuscript: the GB1 coding sequence from Olson et al. [17]; the MPRA backbone, TFBS sequences, and inert background sequence from Melnikov et al. [3] and Georgakopoulos-Soares et al. [56]; the SMN2 exon 7 genomic context from GRCh38.p13 (RefSeq assembly accession GCF_000001405.39); the human 5' splice-site position weight matrix from Wong et al. [45]; MaxEntScan from Yeo and Burge [46]; and the pretrained SpliceAI model ensemble from Jaganathan et al. [44].

Code availability

Project name: PoolParty. Project home page: https://github.com/jbkinney/poolparty-statetracker. Documentation: https://poolparty.readthedocs.io. Archived version: https://doi.org/10.5281/zenodo.19445098. Operating system(s): Platform independent. Programming language: Python (≥ 3.10). Other requirements: NumPy (≥ 1.20), pandas (≥ 1.3), beartype (≥ 0.22.9), pyfaidx (≥ 0.8.1), typing_extensions (≥ 4.0). The companion StateTracker package is installed automatically alongside PoolParty. License: MIT License. Any restrictions to use by non-academics: None.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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Supplemental Information for "PoolParty: streamlined design of DNA sequence libraries in Python"

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Supplemental Methods

Benchmarking analysis

Figure S2 and Table S1 present a benchmarking study of PoolParty using the three library designs featured in Figs. 2, 3, and 4 of the main text. Benchmarking experiments were run under Ubuntu 22.04.3 LTS in Windows Subsystem for Linux 2 on an Intel Core Ultra 9 185H workstation with 31 GB RAM allocated to WSL2, using CPython 3.12.2, PoolParty 0.1.0, and StateTracker 0.1.0. Timings are single-threaded wall-clock measurements of `generate_library`, reported as mean ± SD over five replicate runs. Peak memory is the maximum resident set size reported by `getrusage`, with each data point measured in a fresh process.

Tool comparisons

A literature search identified 13 tools with substantial functional overlap with PoolParty; these are listed in Table S2. Table S3 compares the specific features and capabilities of the 7 tools most similar in purpose to PoolParty. Of these 7, we judged VaLiAnT [1], MPRAnator [2], and tangermeme [3] to be the closest competitors. MPRAnator is a web application that, as of this writing, is no longer accessible. Figs. S3 and S4 provide explicit comparisons of the code/inputs required to construct identical libraries using PoolParty versus VaLiAnT (Fig. S3) or PoolParty versus tangermeme (Fig. S4). We note that neither VaLiAnT nor tangermeme is capable of generating all three of the libraries featured in main-text Figs. 2, 3, and 4.

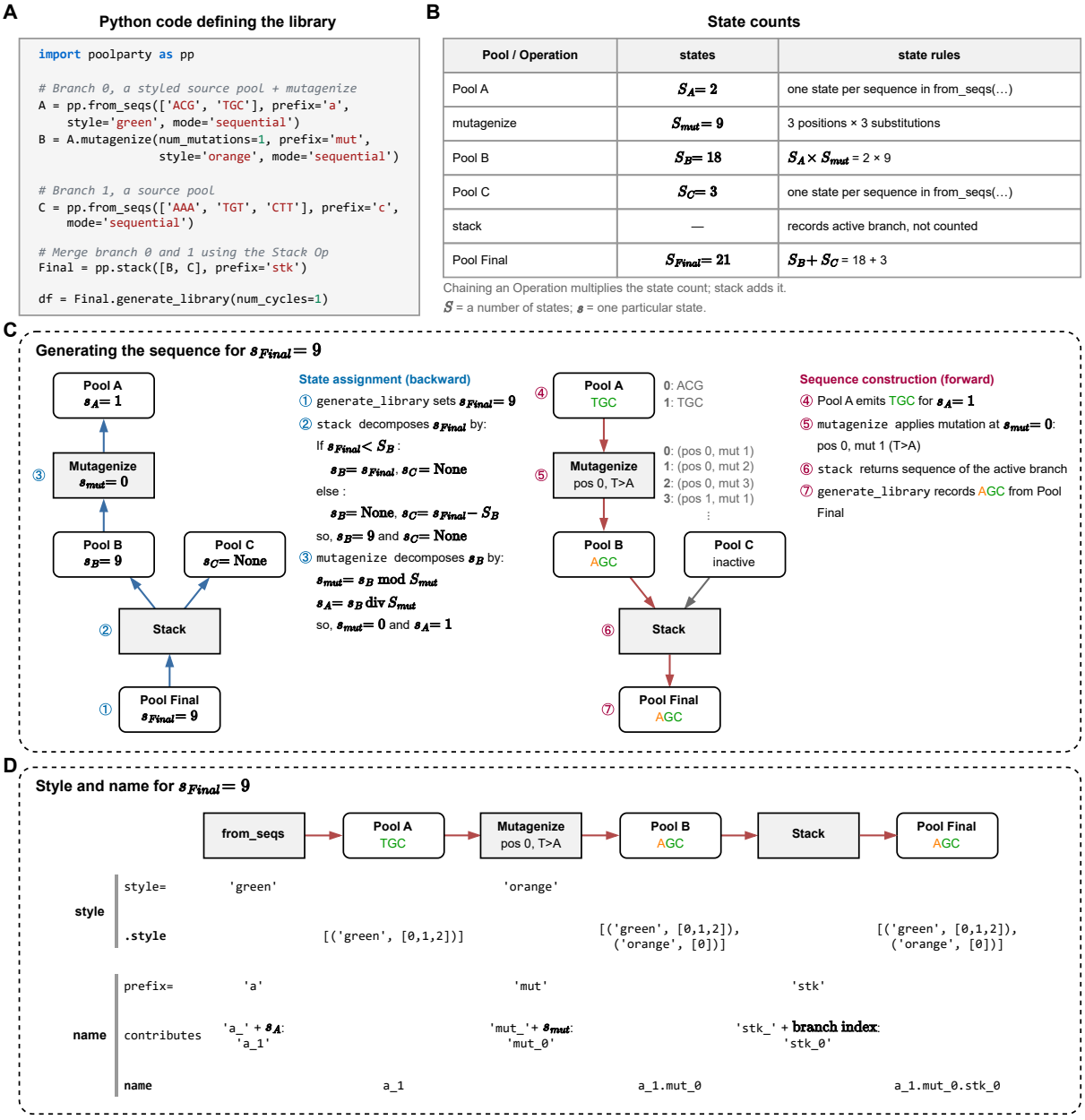


Figure S1. A worked example of sequence generation. (A) Python code defining the DAG in Fig. 1B of the main text. **(B)** Number of states of each Pool and Operation. **(C)** Generating the sequence for $s_{Final} = 9$. Left: state assignment (blue). Right: sequence construction (magenta). Lists beside Pool A and `mutagenize` show which sequence or mutation each state selects. Inactive branches are shown in gray. **(D)** Style and name for the same sequence. Sequences are carried through the DAG as `Seq` objects, which hold the sequence string together with its `.style`. The `style` and `prefix` arguments are supplied to Operations; the resulting `.style` and name are held by Pools. Sequence characters are colored by their `.style` entries; where entries overlap, the last one applies. The styling shown in Panel D was generated by setting `_include_inline_styles=True` in `Final.generate_library()`; this is omitted from the code snippet in panel A.

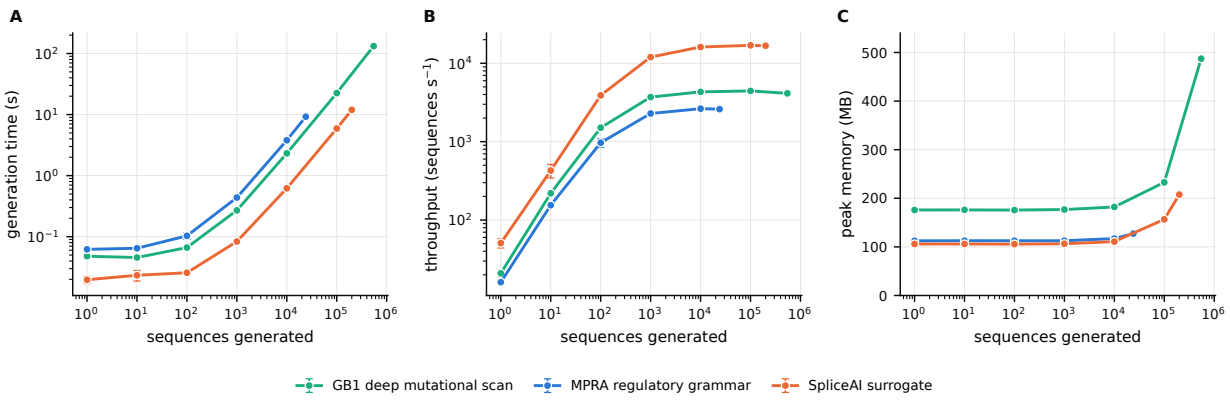
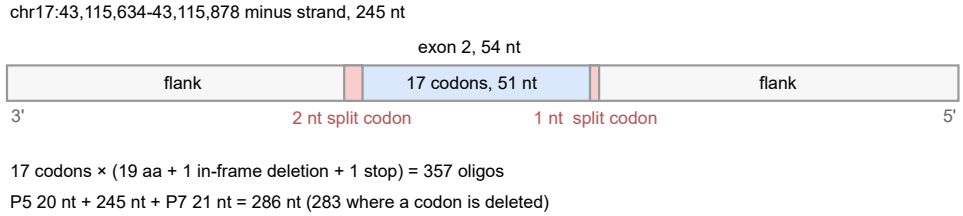


Figure S2. Library generation scales linearly in time and memory. **(A)** Wall-clock generation time, **(B)** throughput, and **(C)** peak resident memory against the number of sequences generated, for the three worked examples. Points are means of 5 replicate runs; error bars show ± 1 SD and are smaller than the plotting symbols at most points. Below $\sim 10^3$ sequences a fixed setup cost dominates, giving the plateau in **(A)** and the rise in **(B)**; above 10^4 sequences throughput is constant, so generation time is linear in library size. Peak memory increases linearly, by 0.52–0.66 kB per sequence above a $\sim$106 MB baseline for the MPRA and SpliceAI libraries; similar behavior is observed for the GB1 library, but with a higher memory baseline.

A The library: BRCA1 exon 2, three mutation types, P5/P7 adaptors



17 codons × (19 aa + 1 in-frame deletion + 1 stop) = 357 oligos

P5 20 nt + 245 nt + P7 21 nt = 286 nt (283 where a codon is deleted)

B PoolParty

```
import poolparty as pp

ADAPTOR_5 = "AATGATACGGCGACCACCGA"
ADAPTOR_3 = "TCGTATGCCGTCTTCTGCTTG"

# Extract region and annotate the target CDS
orf = pp.from_fasta(
    "chr17.fa", ("chr17", 43115633, 43115878, "-")
).annotate_orf(region_name="cds", extent=(100, 151), frame=1)

# Amino-acid substitutions
aa = orf.mutagenize_orf(
    region="cds", num_mutations=1,
    mutation_type="missense_only_first", mode="sequential")

# In-frame codon deletion scan
inframe = orf.deletion_scan_orf(
    deletion_codons=1, deletion_marker=None,
    region="cds", mode="sequential")

# Stop codon scan
stop = orf.insertion_scan_orf(
    "TGA", region="cds", replace=True, mode="sequential")

# Stack three pools, then add the adaptors
library = pp.join([ADAPTOR_5, pp.stack([aa, inframe, stop]),
                   ADAPTOR_3])

library_df = library.generate_library()

Inputs: 1 FASTA
Outputs: oligos, with optional design card parameters that
         produced them
```

C VaLiAnT

```
# Where to mutate and how
targeton.tsv
   ref_chr       chr17
   ref_strand    -
   ref_start     43115634
   ref_end       43115878
   r2_start      43115726
   r2_end        43115779
   ext_vector    "25,25"
   action_vector "(),(aa,inframe,stop),()"

# The transcript model, 51 records
ENST00000357654.9.gtf
   ...
   CDS 43115726 43115779 . - 1
   ...

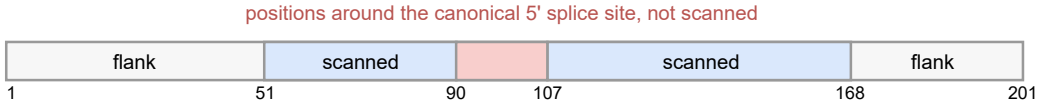
# The reference genome
chr17.fa 81 MB

# Adaptors, strand, and annotation as flags
$ valiant sge targeton.tsv ../ref/chr17.fa out \
      'homo sapiens' 'GRCh38' \
      --revcomp-minus-strand \
      --adaptor-5 ... --adaptor-3 ... \
      --gff ENST00000357654.9.gtf

Inputs: 3 files
Outputs: oligos, genomic coordinates,
         consequence annotation, and a VCF
```

Figure S3. The same BRCA1 exon 2 variant library in PoolParty and VaLiAnT. **(A)** For each of the 17 complete codons in BRCA1 exon 2, the library contains substitutions to the other 19 amino acids, one stop-codon substitution, and one in-frame codon deletion. This gives 357 oligos. The split codons at the exon boundaries are left unchanged, and P5 and P7 adaptors are added to each sequence. **(B)** PoolParty implementation. The reference sequence is loaded from a FASTA file, and the user supplies the extent and reading frame of the 17 complete codons. Three ORF-aware Operations generate the amino-acid, stop-codon, and deletion scans. **(C)** VaLiAnT 4.0.0 implementation. VaLiAnT takes a coordinate and action table, a GTF annotation, and the reference chromosome as input. VaLiAnT gets the CDS phase from the GTF; in PoolParty, the user supplies the ORF extent and frame. The PoolParty code additionally calls `pp.set_genetic_code(valiant_codon_preference())` to match VaLiAnT's arginine codon choice; this setup line is omitted from panel B. When configured to use the same codon usage table, PoolParty and VaLiAnT gave exactly the same set of 357 sequences.

A The library: SMN2 exon 7, 2,000 cryptic 5'ss at 100 positions



2,000 cryptic 5' splice sites × 100 positions = 200,000 sequences

```
SCAN_POSITIONS = list(range(51,90)) + list(range(107,168))
```

B PoolParty

```python
import poolparty as pp

# The 2,000 cryptic sites, as a pool
cryptic_pool = pp.from_seqs(
    cryptic_sites, mode="sequential")

# Replace one site at each position
library = pp.from_seq(target_region)\
    .replacement_scan(
       cryptic_pool,
       positions=SCAN_POSITIONS,
       mode="sequential")

# Final library, raw sequence strings
library_df = library.generate_library()
```

C tangermeme

```python
import torch
from tangermeme.ersatz import substitute
from tangermeme.utils import one_hot_encode

# The 2,000 cryptic sites, as a tensor
cryptic_ohe = torch.stack(
    [one_hot_encode(site)
     for site in cryptic_sites])

# One copy of the region per cryptic site
target_ohe = (one_hot_encode(target_region)
              .unsqueeze(0)
              .expand(len(cryptic_sites), -1, -1))

# Substitute at each position,
# Final library, one-hot encoded tensor
library_ohe = torch.stack(
    [substitute(target_ohe, cryptic_ohe, start=p)
     for p in SCAN_POSITIONS])
```

Figure S4. The same cryptic splice-site perturbation library in PoolParty and tangermeme. **(A)** The library contains 2,000 cryptic 5′ss 9-mers, each substituted at 100 positions across a 201-bp region containing the canonical 5′ss of SMN2 exon 7, giving 200,000 sequences. Positions immediately around the canonical site are excluded. This is the GT arm of the library in main-text Fig. 4; the matched control arm is built the same way. **(B)** PoolParty implementation. The 2,000 sites form a Pool, and `replacement_scan` substitutes them at the 100 positions. The generated sequences are returned as strings in a table. **(C)** tangermeme 1.4.1 implementation. The target region and cryptic sites are one-hot encoded. Substitutions are applied in one batch per position, and the final output is a tensor. After decoding this tensor, the two implementations gave the same set of 200,000 sequences.

Table S1. Runtime and memory for the three worked examples. Each example was generated at the size shown using a single core. Generation times are means ± SD over five replicate runs. Peak memory is the maximum resident set size and includes the fixed memory used by Python and the example DAG. DAG construction does not depend on library size and is reported separately. The SpliceAI library comprises two matched 200,000-sequence pools (cryptic GT and disrupted GA) generated separately; values are for one pool, and the full library takes twice as long.

Example	Library size	Time (s)	Sequence/s	Memory (MB)	DAG (s)
GB1 deep mutational scan	547,230	132.24 ± 5.76	4,138	487	0.22
MPRA regulatory grammar	24,000	9.22 ± 0.22	2,602	127	0.06
SpliceAI surrogate	200,000	11.93 ± 0.27	16,758	208	0.05

Table S2. Tools for designing DNA sequence libraries and for related sequence-design problems. The final two rows group related tools; their members are named and cited individually.

Tool	Purpose / availability	Key features	Output
MPRA Design Tools [4]	Design MPRA libraries and estimate statistical power. R package (GitHub).	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates
MPRAnator [2]	Design MPRA libraries of motif arrangements and sequence variants. Web service.	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Barcoded library of sequences (FASTA)
Mutation Maker [5]	Design mutagenic oligos for building variant libraries in the laboratory. Web application (Docker); REST API.	Degenerate codons at user-listed sites; several substitutions combined in one reaction; de novo gene assembly with per-site variant frequencies	Primers, or gene fragments with the proportion of each variant to pool
Oligopool Calculator [6]	Add barcodes, primers, and spacers to a supplied variant set, and analyze the sequencing readout. Python package (PyPI); command line.	Constraint-aware barcodes, primers, and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready constructs; variant counts from sequencing reads
PoolParty	Design libraries that combine multiple variant types for laboratory assays or for probing genomic AI models. Python package (PyPI).	Libraries specified as a directed acyclic graph of Operations and Pools, allowing different mutagenesis schemes to be combined; sequences generated only on request	Library of sequences, each paired with a record of how it was constructed
tangermeme [3]	Generate perturbed sequence sets to probe cis-regulatory logic in deep learning models. Python package (PyPI).	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences, model predictions, or attribution scores
VaLiAnT [1]	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning. Command-line tool (source, Docker).	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation
General-purpose toolkits *(Biopython, pydna, SeqPro)* [7–9]	Manipulate sequences, simulate cloning, prepare arrays for modeling. Python packages (PyPI).	Parsing, transformation, and file-format support	Transformed sequences
Sequence optimization tools *(CodonGenie, DNA Chisel, ledidi)* [10–12]	Optimize individual sequences for codon usage, synthesis constraints, or a model's objective. Python packages (PyPI); web services.	Degenerate codons covering a target residue set; hard constraints and scored objectives combined in one problem; gradient-based editing guided by a predictive model	An optimized or edited sequence, or a degenerate codon

Table S3. Capabilities of tools for designing DNA sequence libraries and for related sequence-design problems. ● denotes a capability the tool provides; ◐ one provided with restrictions; ○ one the tool does not provide. Capabilities are counted only where the tool provides a dedicated operation, parameter, or mode.

	DNA Chisel	MPRA Design Tools	MPRAnator	Mutation Maker	Oligopool Calculator	**PoolParty**	tangermeme	VaLiAnT
Library Specification								
Library object	○	◐	○	◐	◐	●	◐	○
Composable operations	◐	○	◐	○	●	●	◐	○
Mixed variant types in one library	○	●	●	◐	○	●	○	●
Variant Generation								
Saturation mutagenesis	○	○	○	○	◐	●	●	●
Randomly sampled variants	●	◐	●	○	○	●	○	○
Pairwise and higher-order variants	●	○	●	●	◐	●	◐	○
Model-guided variants	●	○	○	○	○	◐	●	○
Variant Types								
Codon / amino-acid substitutions	●	○	○	◐	○	●	○	●
Insertions and deletions	○	●	◐	○	○	●	●	●
Combinatorial multi-motif placement	○	○	●	○	○	●	●	○
Recombination	○	○	○	○	○	●	○	○
Shuffling	○	○	◐	○	○	●	●	○
Physical Construction								
Synthesis-constraint checking	●	◐	◐	○	●	◐	○	◐
Constraint-based optimization	●	◐	◐	●	●	○	○	○
Primer design	◐	○	○	●	●	○	○	○
Metadata and Inspection								
Per-sequence construction records	●	●	◐	●	●	●	○	●
Automatic naming	○	◐	●	●	◐	●	◐	●
Sequence styling	○	○	◐	○	○	●	○	○
Genomic Integration								
Genome coordinates	○	◐	●	○	○	◐	◐	●
Transcript / annotation aware	○	○	○	○	○	○	○	●

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